

ANCHORAGE BRYANT AAF, AK, USA

WMO: 702700

Lat: 61.266N

Lon: 149.653W

Elev: 118

StdP: 99.91

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 00489

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.8	-24.5	-34.0	0.2	-27.0	-29.4	0.3	-24.0	11.3	4.1	9.0	2.3	0.7	30	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	24.3	16.0	22.5	15.2	20.7	14.3	16.8	23.0	15.7	21.5	14.8	19.7	2.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.8	10.0	18.9	12.9	9.4	17.1	12.4	9.1	16.7	47.1	22.7	44.1	21.9	41.4	19.7	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	6.8	5.3	-30.4	27.5	4.0	1.5	-33.3	28.6	-35.6	29.5	-37.8	30.3	-40.8	31.4
			-30.5	18.3	3.9	1.0	-33.4	19.0	-35.7	19.6	-37.9	20.2	-40.7	21.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-9.4	-9.0	-3.7	2.7	8.8	13.3	15.3	13.6	8.3	1.6	-6.3	-7.6
	DBStd	10.29	7.69	7.04	5.70	3.74	2.68	2.44	2.08	2.37	3.83	4.73	6.77	6.57
	HDD10.0	3206	602	532	425	219	55	1	0	2	73	263	489	546
	HDD18.3	5833	860	765	683	469	294	152	98	146	301	520	739	804
	CDD10.0	419	0	0	0	0	19	100	163	114	21	1	0	0
	CDD18.3	6	0	0	0	0	0	2	3	1	0	0	0	0
	CDH23.3	102	0	0	0	0	1	29	62	11	0	0	0	0
	CDH26.7	13	0	0	0	0	0	3	11	0	0	0	0	0
Wind	WSAvg	1.3	1.2	1.2	1.4	1.6	1.8	1.5	1.2	1.2	1.4	1.1	1.1	1.3
Precipitation	PrecAvg	457	27	26	21	13	18	28	48	70	85	48	35	38
	PrecMax	591	55	60	51	46	42	66	92	170	176	118	76	69
	PrecMin	339	5	3	3	3	5	8	15	28	25	17	2	3
	PrecStd	62	15	15	11	9	9	14	21	36	41	22	19	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	6.7	8.3	13.8	22.5	26.6	27.9	24.9	18.4	13.6	8.0	8.2
		MCWB	3.1	3.1	4.0	7.4	12.9	16.2	17.7	16.8	12.4	8.5	3.9	3.9
	2%	DB	5.5	4.8	6.5	11.7	19.2	23.5	25.0	23.0	16.6	11.4	6.0	6.4
		MCWB	2.4	1.7	3.0	6.3	11.5	14.8	16.4	16.0	11.6	7.2	2.8	2.9
	5%	DB	3.7	2.8	5.2	10.3	17.0	21.3	22.9	21.0	15.2	9.8	3.7	3.9
		MCWB	1.1	0.7	2.2	5.3	10.5	14.1	15.6	14.7	10.9	6.4	2.1	1.7
	10%	DB	1.0	0.4	3.9	9.0	15.0	19.1	21.1	18.9	13.8	8.3	2.0	0.8
		MCWB	-0.2	-0.7	1.1	4.6	9.3	12.8	14.7	13.5	10.3	5.8	1.0	-0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.7	4.6	8.3	13.5	17.2	18.7	18.2	13.5	9.4	4.4	4.1
		MCDB	6.4	6.3	7.4	12.8	21.2	24.0	28.0	23.7	16.3	12.1	6.8	7.2
	2%	WB	2.4	2.0	3.5	6.8	12.0	15.6	16.9	16.6	12.3	7.9	3.3	2.9
		MCDB	5.2	4.4	6.0	11.2	18.4	22.8	23.8	21.8	15.1	10.2	5.3	5.8
	5%	WB	1.2	0.7	2.3	5.7	10.7	14.3	15.8	15.3	11.6	7.0	2.2	1.5
		MCDB	3.3	2.5	4.8	9.8	16.0	20.2	21.8	20.2	14.0	9.1	3.7	3.5
	10%	WB	-0.2	-0.8	1.4	4.8	9.7	13.1	15.1	14.3	10.8	6.2	0.9	-0.1
		MCDB	1.3	0.6	3.6	8.5	14.4	18.8	20.3	17.9	13.1	8.1	1.9	0.8
Mean Daily Temperature Range	5% DB	MDBR	7.9	10.1	10.5	10.3	10.1	9.7	9.2	8.7	8.0	7.6	7.5	7.3
		MCDBR	8.7	9.3	9.2	12.6	14.2	14.4	14.5	13.0	8.9	8.5	7.7	8.6
		MCWBR	6.5	7.0	6.5	8.3	8.4	8.0	7.6	6.8	5.2	5.3	5.4	5.9
	5% WB	MCDBR	8.3	8.4	8.9	11.3	13.1	12.9	12.6	11.1	7.7	7.3	7.2	8.0
		MCWBR	6.4	6.5	6.6	7.8	7.9	7.6	7.1	6.1	5.0	4.9	5.2	6.0
Clear-Sky Solar Irradiance	taub	0.229	0.282	0.302	0.340	0.359	0.354	0.366	0.350	0.312	0.293	0.238	N/A	
	taud	2.121	2.229	2.260	2.244	2.312	2.411	2.413	2.464	2.557	2.453	2.113	N/A	
	Ebn at Noon	508	697	817	849	861	873	851	838	818	698	474	N/A	
	Edh at Noon	37	65	89	111	113	104	102	89	68	53	36	N/A	
All-Sky Solar Radiation	RadAvg	0.29	0.98	2.59	4.30	5.18	5.32	4.75	3.76	2.44	1.13	0.42	0.15	
	RadStd	0.04	0.13	0.27	0.37	0.44	0.33	0.38	0.40	0.33	0.13	0.06	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (25 neighbors)	N/A	+1.98	+1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page